Sriya Choudary Yalavarthy

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EDUCATION

University of Massachusetts Amherst, Amherst, MA — Aug 2023 – May 2025
M.S. in Computer Science — GPA: 3.71
Coursework: Machine learning, Systems for Data Science, Neural Networks, Artificial Intelligence, Algorithms for Data Science.

SRM Institute of Science and Technology, Chennai, India — July 2019 – May 2023
B.S. in Computer Science, Minor in AI/ML — GPA: 3.9

SKILLS

Languages: Python, Java, C++, R, SQL, JavaScript
Machine Learning & AI: Scikit-learn,Pandas,Plotly, XGBoost, PySpark, LightGBM, TensorFlow, PyTorch, MLflow, SHAP
Generative AI & NLP: Hugging Face Transformers, spaCy, GPT-3.5, BERT, OpenAI API, TextBlob
MLOps & Deployment: Docker, Kubernetes, Airflow, Flask, FastAPI, Git, CI/CD, Streamlit
Cloud & Visualization: AWS (EC2, S3, SageMaker),Azure, GCP, Tableau, Power BI, Matplotlib, Seaborn, Databricks, Excel

EXPERIENCE

University Of Massachusetts, USA — Aug 2025 – Present
Data Analyst CICStep Participant
- Processing air quality datasets by architecting Snowflake multi-warehouse ETL pipeline with JSON ingestion, reducing query processing time by 60% through optimized schema design.
- Transforming raw environmental data using separate Snowflake schemas for loading, cleaning, and publishing stages, ensuring data quality standards and governance compliance.
- Optimizing cloud data warehouse performance by implementing staged processing across multiple Snowflake environments, enabling scalable air quality analytics for research initiatives.

Sri Sai Speciality Dental Hospital, India — May 2024 – Aug 2024
Data Analyst Intern
- Analyzed patient health and operational data using Python, SQL, and Excel to optimize patient scheduling and resource allocation, improving efficiency by 20%.
- Developed interactive dashboards in Power BI and Tableau to monitor admissions, treatment outcomes, and resource utilization, reducing manual reporting effort by 35%.
- Automated ETL workflows and recurring reports with Python and SQL to cut data processing time from 3 hours to 20 minutes while ensuring 100% reporting accuracy.

MedTourEasy, India — Jan 2023 – Jul 2023
Data Science Intern
- Designed and deployed a production-grade machine learning pipeline to process 10K+ daily user activity records, enabling pattern recognition for 7 cardiovascular health indicators with 92% model accuracy.
- Developed predictive models using scikit-learn and gradient boosting algorithms, increasing early disease detection by 15% with an AUC-ROC of 0.87 and 5-fold cross-validation.
- Performed statistical analysis using statsmodels to extract 5 key risk factors from 12+ features and created correlation matrices.

HighRadius, India — Jan 2022 – May 2022
Data Science Intern
- Developed and deployed a machine learning model to forecast payment due dates using 5+ years of historical data, reducing manual tracking by 40 hours/month and improving cash flow accuracy by 20%.
- Integrated predictive analytics into a React-based dashboard, enabling real-time forecasting for 100+ recurring transactions/month with 500ms latency, reducing decision-making time.

PROJECTS

AI-Powered Murder Mystery Investigation Webapp
- Implemented Retrieval-Augmented Generation (RAG) with HuggingFace embeddings to create contextually aware AI character responses that maintain consistency across conversations.
- Engineered vector similarity search with cosine similarity algorithms to ensure uniqueness in AI-generated case scenarios, preventing duplicate content.
- Designed an embedding-based content retrieval system for character memory, allowing AI suspects to reference past interactions and maintain persistent personalities during interrogations.

Financial Fraud Detection System
- Designed and deployed a real-time fraud detection model using ensemble machine learning algorithms to enhance fraud identification accuracy and reliability.
- Engineered a high-dimensional feature set using transactional and behavioral signals via Kafka and Redis to support low-latency predictions.
- Developed an end-to-end MLOps pipeline with automated retraining and integrated SHAP-based model explainability to support transparent and scalable fraud detection in production systems.

Generative AI Resume Screening Assistant
- Developed a transformer-based NLP pipeline using BERT embeddings to extract skills and match resumes to job descriptions with high precision.
- Integrated GPT-3.5 to auto-generate personalized resume feedback in real-time, simulating recruiter insights at scale.
- Scaled the pipeline architecture to handle multiple document types and integrated real-time processing capabilities for enterprise deployment.